COMP102 Lecture 6: Binary Search Trees

Search, Ordering Invariants, Height, and AVL Trees

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THE UM6P VANGUARD CENTER

#Empowering Minds.

Plan

1. Learning Outcomes
2. From Binary Trees to BSTs
3. Searching
4. Insertion and Sorted Traversal
5. Height and Complexity
6. Self-Balancing Trees
7. AVL Trees
8. Wrap-up

Learning Outcomes

Learning outcomes

By the end of this lecture, students should be able to:

- define the **binary search tree** property;
- check whether a binary tree is a valid BST;
- trace **search** and **insertion** in a BST;
- explain why **inorder traversal** of a BST returns sorted values;
- connect BST performance to the **height** of the tree;
- explain how bad insertion orders create **degenerate** BSTs;
- define the **AVL balance factor**;
- recognize the four AVL imbalance cases and the rotations used to repair them.

Last lecture: a binary tree was a **shape**. This lecture adds an **ordering invariant** so the shape can support efficient search.

From Binary Trees to BSTs

Continuity from last lecture

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Binary tree

A node may have at most two children: **left** and **right**.

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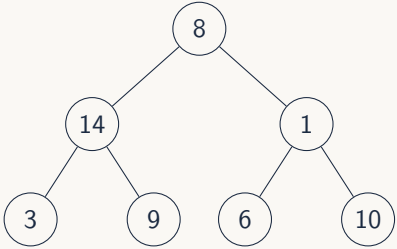

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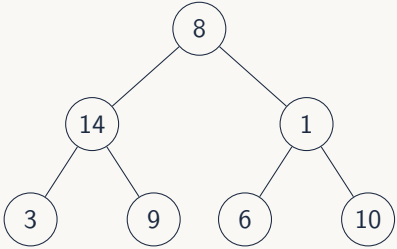

A normal binary tree gives structure, but not search direction.

The binary search tree rule

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BST property

For every node:

$$\text{all values in left subtree} < \text{node value} < \text{all values in right subtree}$$

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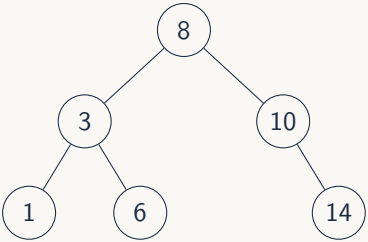

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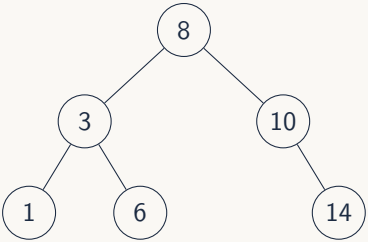

Same node class. Stronger meaning.

Same class, new invariant

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Python Node

```python
class Node:
    def __init__(self, value):
        self.value = value
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Example 2.3. What changed?

The fields are identical to last lecture. The new part is a rule about where values are allowed to appear.

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Invariant

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An invariant is a property that must remain true after every operation.

Example 2.5. What changed?

The fields are identical to last lecture. The new part is a rule about where values are allowed to appear.

Valid or invalid?

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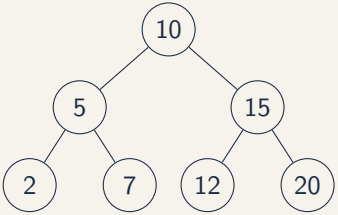

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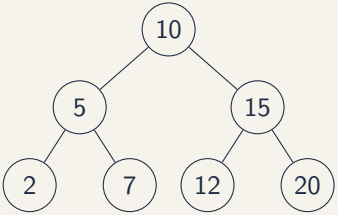

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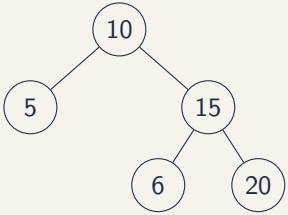

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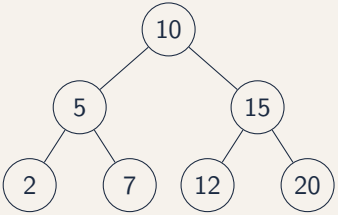

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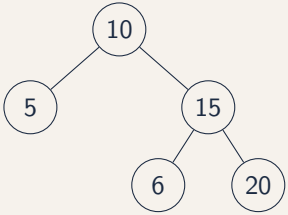

The value 6 is in the right subtree of 10, so it must be greater than 10. It is not. Therefore the tree is not a BST.

Searching

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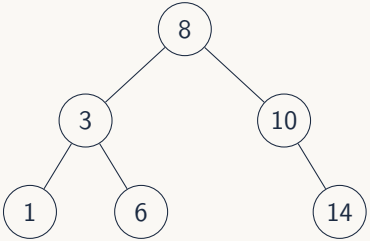

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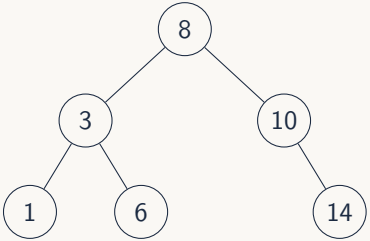

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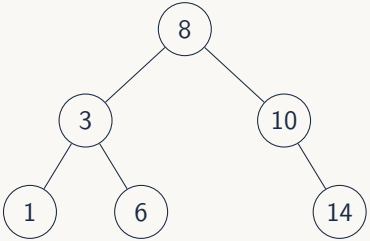

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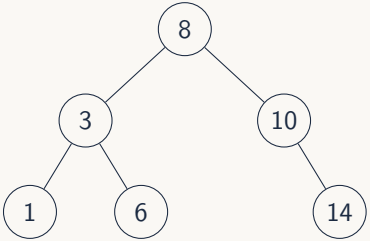

Remark 3.5. Key difference

In an ordinary binary tree, we may have to search both sides. In a BST, one side is ruled out.

BST search in Python

BST search in Python

Recursive search

```python
def search(root, target):
    if root is None:
        return False

    if target == root.value:
        return True
    elif target < root.value:
        return search(root.left, target)
    else:
        return search(root.right, target)
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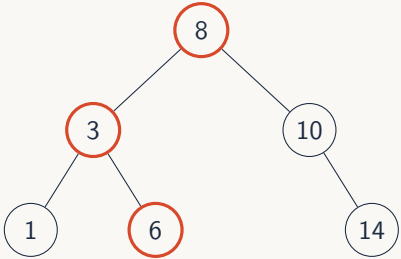
The code is short because the BST invariant does most of the work.

Trace a search

Search for **6**.



- Compare with 8: $6 < 8$, go left.

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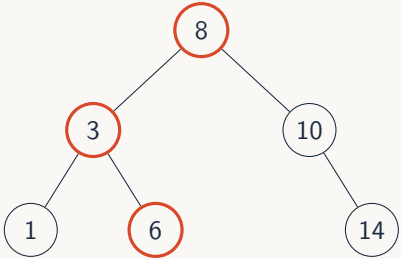

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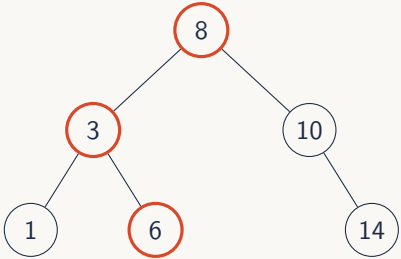

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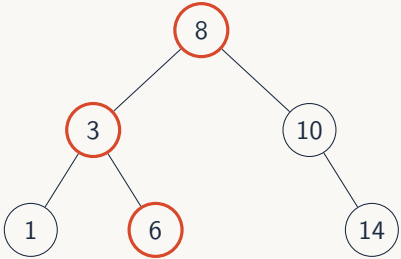

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Path

$8 \rightarrow 3 \rightarrow 6$

Insertion and Sorted Traversal

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Insertion is search for the correct empty position.

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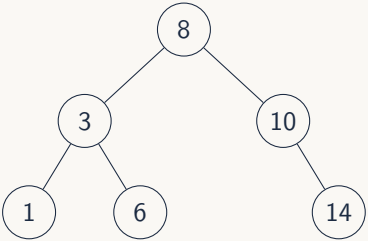
- Start at the root.



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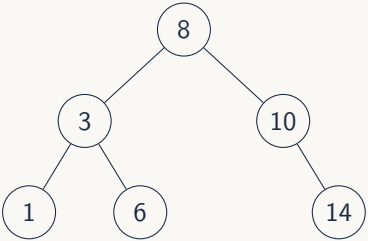

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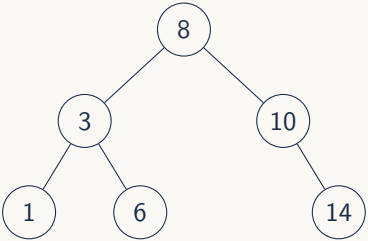

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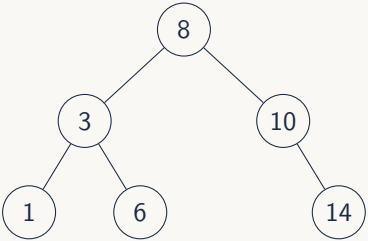

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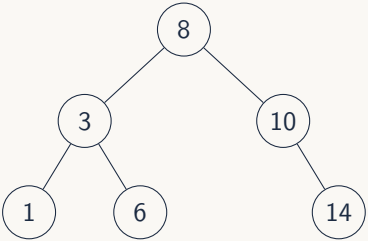

Remark 4.6. For this lecture

We will assume duplicate values are not inserted.

BST insertion in Python

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Recursive insertion

```python
def insert(root, value):
    if root is None:
        return Node(value)

    if value < root.value:
        root.left = insert(root.left, value)
    elif value > root.value:
        root.right = insert(root.right, value)

    return root
```

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Each recursive call moves into the only subtree where the new value is allowed to appear.

Build a BST step by step

Insert the values:

$$8,\ 3,\ 10,\ 1,\ 6,\ 14$$

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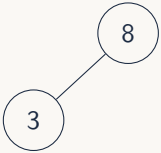

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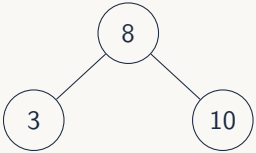

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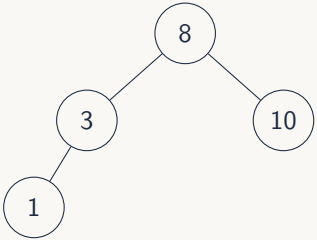

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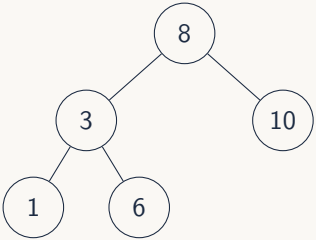

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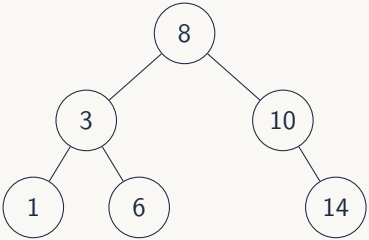

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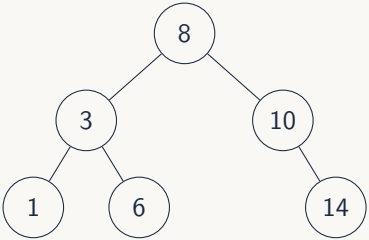

The final shape depends on the insertion order.

Inorder becomes sorted order

From last lecture:

$$\text{inorder} = \text{Left, Node, Right}$$

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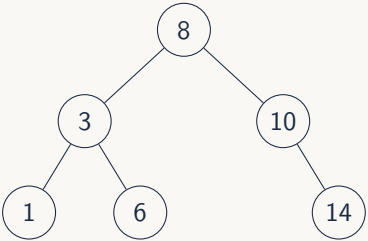

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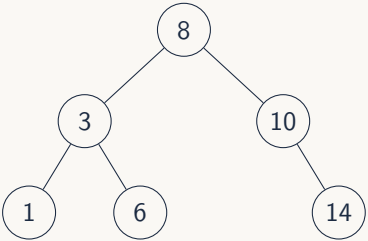

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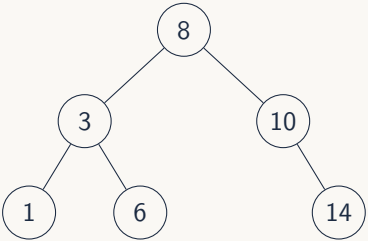

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The traversal did not change. The meaning changed because the tree now has an ordering invariant.

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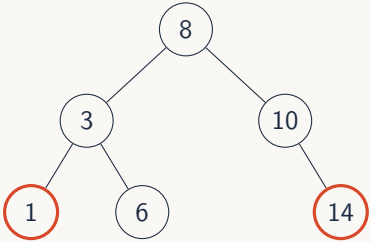
Minimum

Keep following **left** pointers until there is no left child.



Minimum and maximum

The BST rule makes two operations especially simple.

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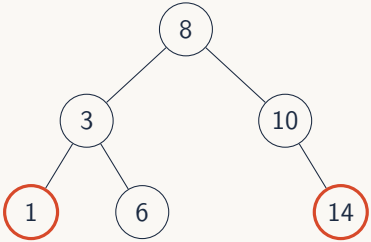

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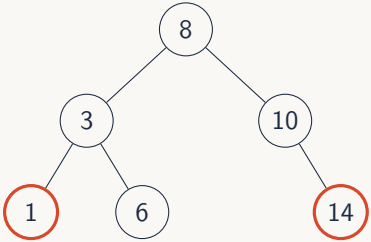


These operations still depend on height: they follow one path from the root.

Height and Complexity

Cost depends on height

Let h be the height of the tree.

Operation	**Cost**
Search	$O(h)$
Insert	$O(h)$
Minimum / maximum	$O(h)$
Inorder traversal	$O(n)$

Cost depends on height

Let h be the height of the tree.

Operation	**Cost**
Search	$O(h)$
Insert	$O(h)$
Minimum / maximum	$O(h)$
Inorder traversal	$O(n)$

Remark 5.2. Main question

How large can h become?

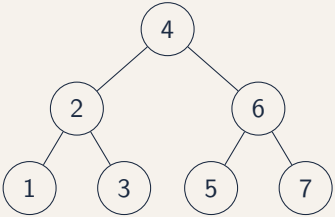
Balanced vs degenerate shape

Same values. Different height.

Balanced vs degenerate shape

Same values. Different height.

Balanced shape

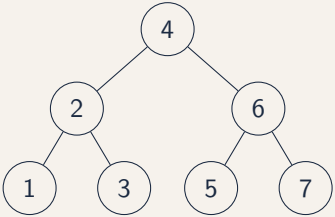


height $= 2$

Balanced vs degenerate shape

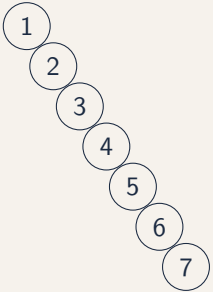
Same values. Different height.

Balanced shape



height $= 2$

Degenerate shape

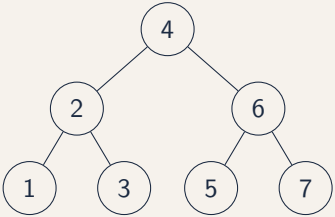


height $= 6$

Balanced vs degenerate shape

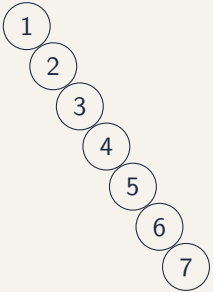
Same values. Different height.

Balanced shape



height $= 2$

Degenerate shape



height $= 6$

The BST rule gives search direction, but it does not guarantee a good shape.

The bad insertion order

Insert sorted values:

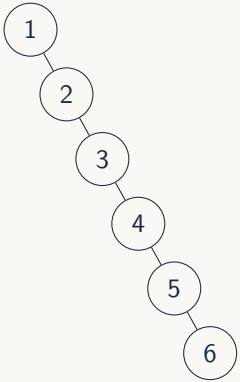
$$1,\ 2,\ 3,\ 4,\ 5,\ 6$$

The bad insertion order

Insert sorted values:

$$1,\ 2,\ 3,\ 4,\ 5,\ 6$$

- Every new value is larger than all previous values.

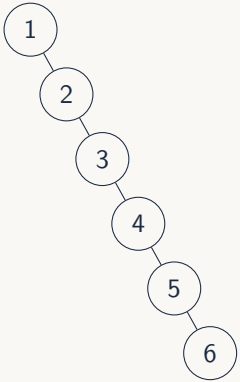


The bad insertion order

Insert sorted values:

$$1,\ 2,\ 3,\ 4,\ 5,\ 6$$

- Every new value is larger than all previous values.
- Every insertion goes to the right.



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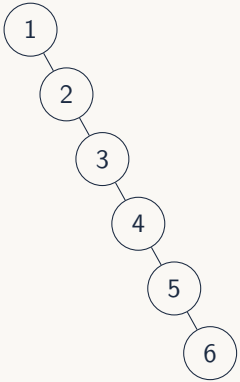
- The BST becomes a linked list in disguise.



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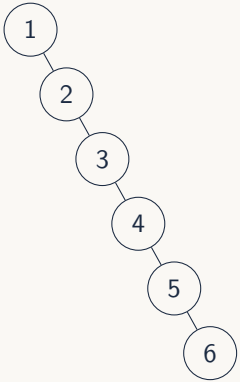

Worst-case BST search is $O(n)$ because the height can become $n - 1$.

Best case, worst case

Shape	**Height**	**Search / insert**
Balanced BST	$O(\log n)$	$O(\log n)$
Degenerate BST	$O(n)$	$O(n)$

Best case, worst case

Shape	**Height**	**Search / insert**
Balanced BST	$O(\log n)$	$O(\log n)$
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A BST is only efficient if the height stays small.

Best case, worst case

Shape	Height	Search / insert
Balanced BST	$O(\log n)$	$O(\log n)$
Degenerate BST	$O(n)$	$O(n)$

A BST is only efficient if the height stays small.

Remark 5.5. Transition

Self-balancing trees add one more promise: after updates, they repair the shape.

Self-Balancing Trees

Self-balancing trees

Self-balancing trees

Problem

A normal BST can become tall depending on insertion order.

Self-balancing trees

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A normal BST can become tall depending on insertion order.

Idea

A self-balancing BST keeps the ordering rule, but also controls the height.

Self-balancing trees

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Examples include:

- AVL trees
- red-black trees
- B-trees

Self-balancing trees

Problem

A normal BST can become tall depending on insertion order.

Idea

A self-balancing BST keeps the ordering rule, but also controls the height.

Examples include:

- AVL trees
- red-black trees
- B-trees

In this lecture, AVL trees are the only concrete example.

AVL Trees

AVL trees: definition

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Balance factor

For any node:

$$\text{balance factor} = \text{height(left subtree)} - \text{height(right subtree)}$$

AVL trees: definition

Balance factor

For any node:

$$\text{balance factor} = \text{height(left subtree)} - \text{height(right subtree)}$$

AVL condition

Every node must have balance factor:

$$-1,\ 0,\ \text{or } 1$$

AVL trees: definition

Balance factor

For any node:

$$\text{balance factor} = \text{height(left subtree)} - \text{height(right subtree)}$$

AVL condition

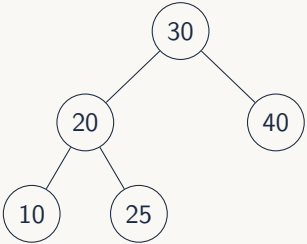
Every node must have balance factor:

$$-1,\ 0,\ \text{or } 1$$

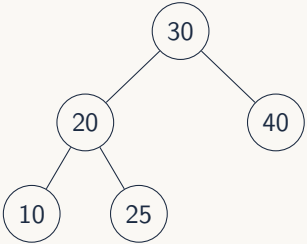
An AVL tree is a BST that refuses to become too left-heavy or too right-heavy at any node.

Compute balance factors

Compute balance factors

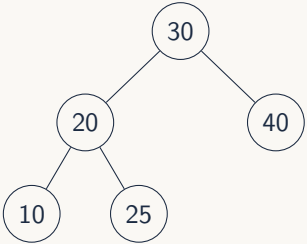


Compute balance factors



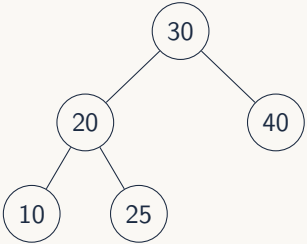
- leaves have height 0;

Compute balance factors



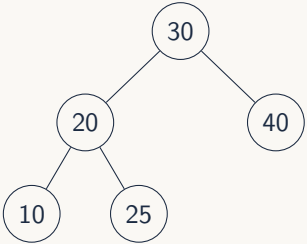
- leaves have height 0;
- node 20 has left height 0 and right height 0;

Compute balance factors



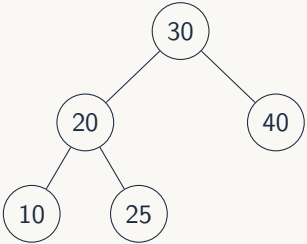
- leaves have height 0;
- node 20 has left height 0 and right height 0;
- root 30 has left height 1 and right height 0;

Compute balance factors



- leaves have height 0;
- node 20 has left height 0 and right height 0;
- root 30 has left height 1 and right height 0;
- balance factor at 30 is $1 - 0 = 1$.

Compute balance factors

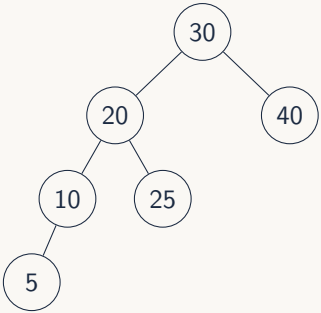


- leaves have height 0;
- node 20 has left height 0 and right height 0;
- root 30 has left height 1 and right height 0;
- balance factor at 30 is $1 - 0 = 1$.

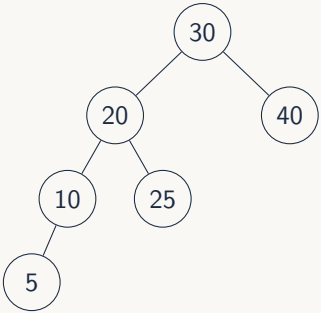
This tree is AVL-balanced because every balance factor is in $\{-1, 0, 1\}$.

When a node is unbalanced

When a node is unbalanced

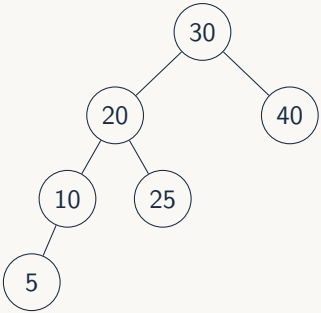


When a node is unbalanced



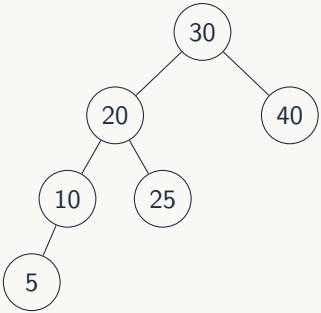
- left subtree of 30 has height 2;

When a node is unbalanced



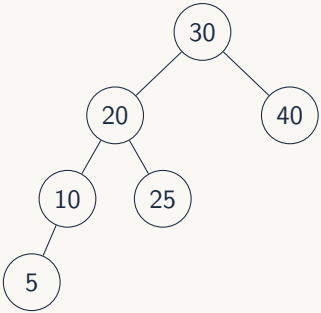
- left subtree of 30 has height 2;
- right subtree of 30 has height 0;

When a node is unbalanced



- left subtree of 30 has height 2;
- right subtree of 30 has height 0;
- balance factor at 30 is $2 - 0 = 2$.

When a node is unbalanced



- left subtree of 30 has height 2;
- right subtree of 30 has height 0;
- balance factor at 30 is $2 - 0 = 2$.

Remark 7.6. Conclusion

Balance factor 2 is not allowed in an AVL tree.

Rotations repair local imbalance

Rotations repair local imbalance

Rotation

A rotation is a local rewrite of a small part of the tree.

Rotations repair local imbalance

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A rotation must preserve:

- the BST ordering property;
- the same set of values;
- a smaller local height.

Rotations repair local imbalance

Rotation

A rotation is a local rewrite of a small part of the tree.

A rotation must preserve:

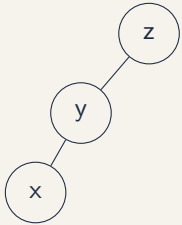
- the BST ordering property;
- the same set of values;
- a smaller local height.

Rotations are not arbitrary rearrangements. They are controlled repairs that keep the inorder order unchanged.

Right rotation: Left-Left case

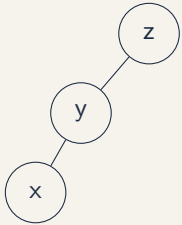
Right rotation: Left-Left case

Before



Right rotation: Left-Left case

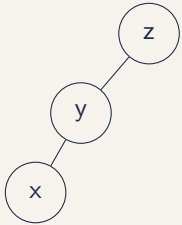
Before



$\Rightarrow$

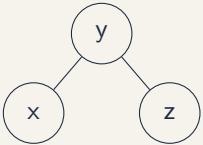
Right rotation: Left-Left case

Before



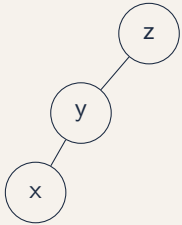
$\Rightarrow$

After



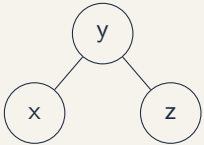
Right rotation: Left-Left case

Before



$\Rightarrow$

After

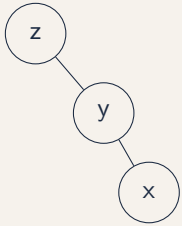


A Left-Left imbalance is repaired by a **right rotation**.

Left rotation: Right-Right case

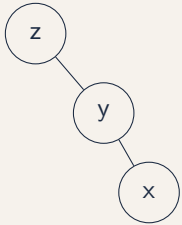
Left rotation: Right-Right case

Before



Left rotation: Right-Right case

Before



$\Rightarrow$

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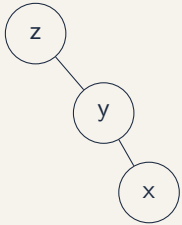

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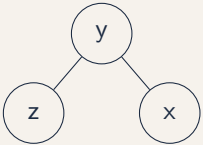
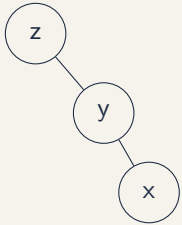
After



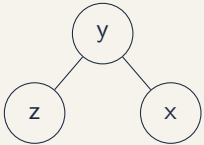
Left rotation: Right-Right case

Before



$\Rightarrow$

After

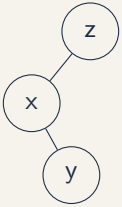


A Right-Right imbalance is repaired by a **left rotation**.

Double rotation: Left-Right case

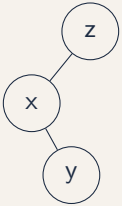
Double rotation: Left-Right case

Left-Right imbalance

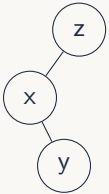


Double rotation: Left-Right case

Left-Right imbalance

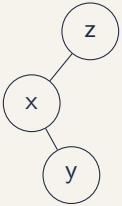


Before

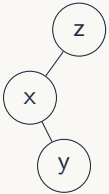


Double rotation: Left-Right case

Left-Right imbalance

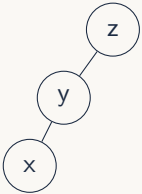


Before



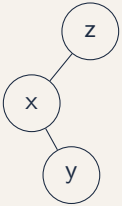
$\Rightarrow$

1. Rotate left at x

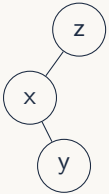


Double rotation: Left-Right case

Left-Right imbalance



Before



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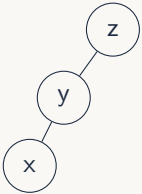
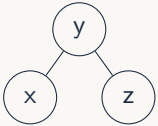


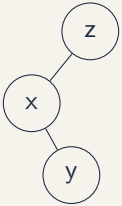
$\Rightarrow$

2. Rotate right at z

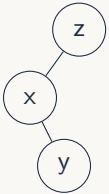


Double rotation: Left-Right case

Left-Right imbalance

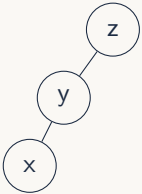


Before



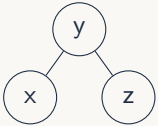
$\Rightarrow$

1. Rotate left at x



$\Rightarrow$

2. Rotate right at z



A Left-Right case is first converted into a Left-Left case, then repaired by a right rotation.

Double rotation: Right-Left case

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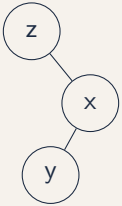
Right-Left imbalance



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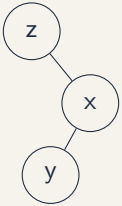

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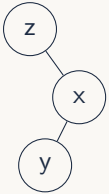

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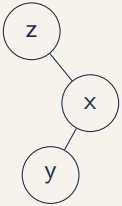

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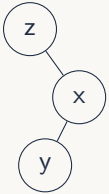

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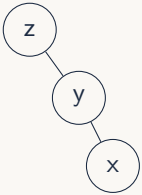

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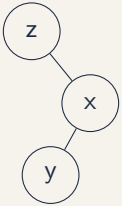

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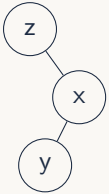

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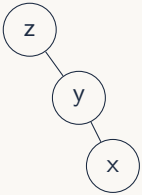

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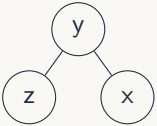

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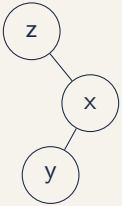

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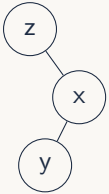

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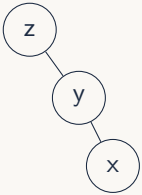

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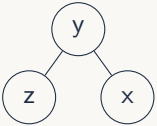

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The four AVL insertion cases

Case	Shape	Repair
Left-Left	left child of left subtree	right rotation
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After insertion, AVL rebalancing repairs the first ancestor whose balance factor becomes invalid.

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1. Insert the value exactly as in a normal BST.
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The BST rule decides *where* the value goes. The AVL rule decides whether the tree needs repair afterward.

Deleting from a BST

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Deletion

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The first two cases are simple. The third case needs a replacement value.

Deletion case 1: leaf node

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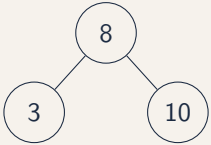
Before



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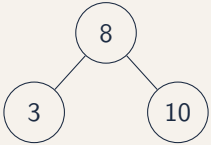

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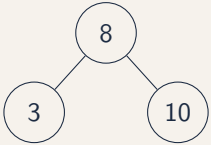
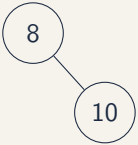


$\Rightarrow$

After deleting 3



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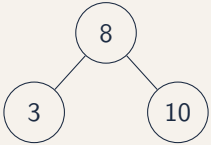

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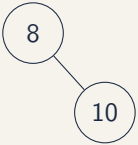

If the node has no children, simply remove it.

Deletion case 2: one child

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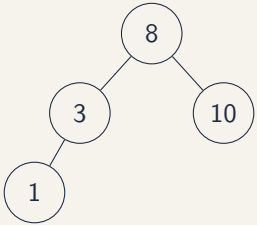
Before



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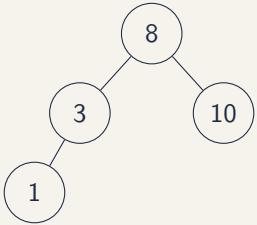

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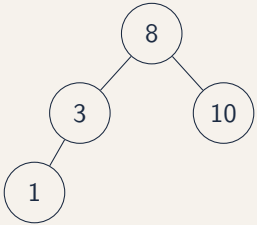

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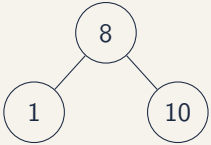

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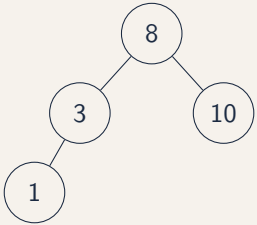

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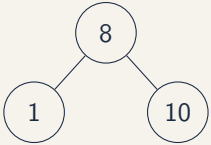

If the node has one child, connect the parent directly to that child.

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If the node has two children, removing it leaves two subtrees that must still fit the BST order.

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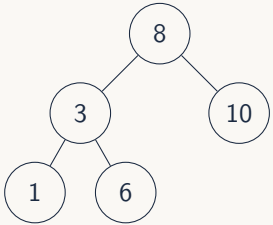

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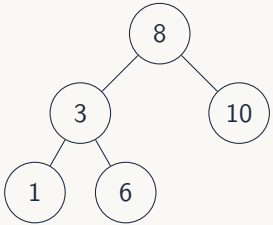

Delete 3

We replace it with either:

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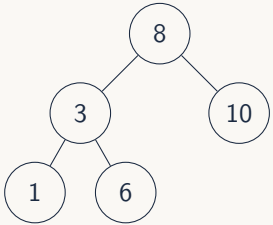

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We replace it with either:

- its **inorder predecessor**; or
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Common choice: use the inorder successor, the smallest value in the right subtree.

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The inorder successor of a node is the next value after it in sorted order.

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To find it:

1. go once to the right;
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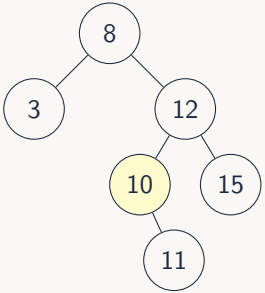

Successor of 8 is 10.

Deleting a node with two children

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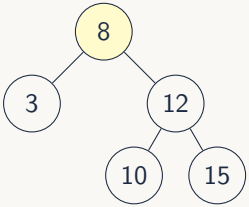
Before



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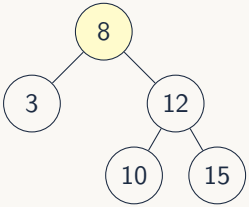
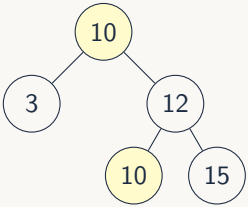


$\Rightarrow$

Copy successor



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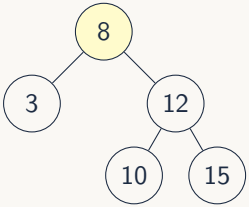

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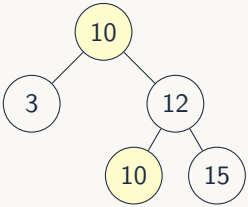

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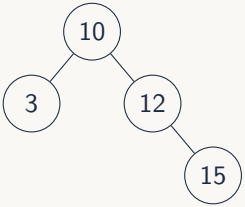
Delete old successor



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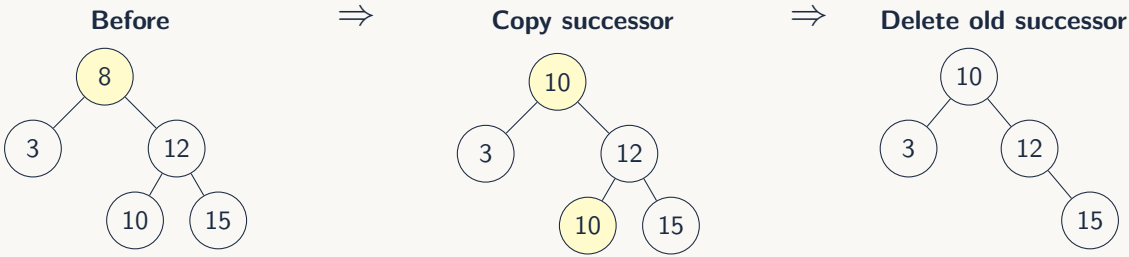

Two-child deletion becomes simpler by replacing the node with its inorder successor, then deleting that successor from its old position.

BST deletion algorithm

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1. If it has no child, return `None`.
2. If it has one child, return that child.
3. If it has two children:
   - 3.1 find the inorder successor;
   - 3.2 copy its value into the current node;
   - 3.3 delete the successor from the right subtree.

BST deletion algorithm

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When the node is found:

1. If it has no child, return `None`.
2. If it has one child, return that child.
3. If it has two children:
   - 3.1 find the inorder successor;
   - 3.2 copy its value into the current node;
   - 3.3 delete the successor from the right subtree.

Deletion preserves the BST property by replacing a removed value with the next valid sorted value.

Wrap-up

Comparison

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Binary tree	no ordering rule	$O(n)$
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The main progression is: structure $\rightarrow$ order $\rightarrow$ balance.

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A binary search tree gives us direction. A self-balancing tree keeps that direction efficient.